

FTM iMCQ 10

Fall 2025

12-GNET questions

A 2-year-old boy is brought to the clinic by his parents because of poor cognitive development for his age. Physical examination shows microcephaly and epicanthal folds. Auscultation of the heart shows a holosystolic murmur with a thrill. During examination, a high-pitched cry like a cat is heard. Which of the following is the most likely chromosomal abnormality in this patient?

- | | |
|---------------------------------|-----|
| A. Trisomy | 17% |
| B. Monosomy of a sex chromosome | 17% |
| C. Chromosomal deletion | 17% |
| D. Chromosomal duplication | 17% |
| E. Chromosomal translocation | 17% |
| F. Triploidy | 17% |

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An 8-year-old boy is brought to the physician by his adoptive parents because of an 8-month history of progressive swelling of the jaw. Laboratory analysis by FISH of cells isolated from tumor tissue shows a translocation between chromosomes 8 and 14. The tumor cells have elevated levels of an oncoprotein. Which of the following best describes the function of this protein?

- A. Has tyrosine kinase activity 17%
- B. Functions as a growth factor receptor 17%
- C. Activates the G1-S checkpoint 17%
- D. Required for DNA repair 17%
- E. Transcription factor 17%
- F. Activates apoptosis 17%

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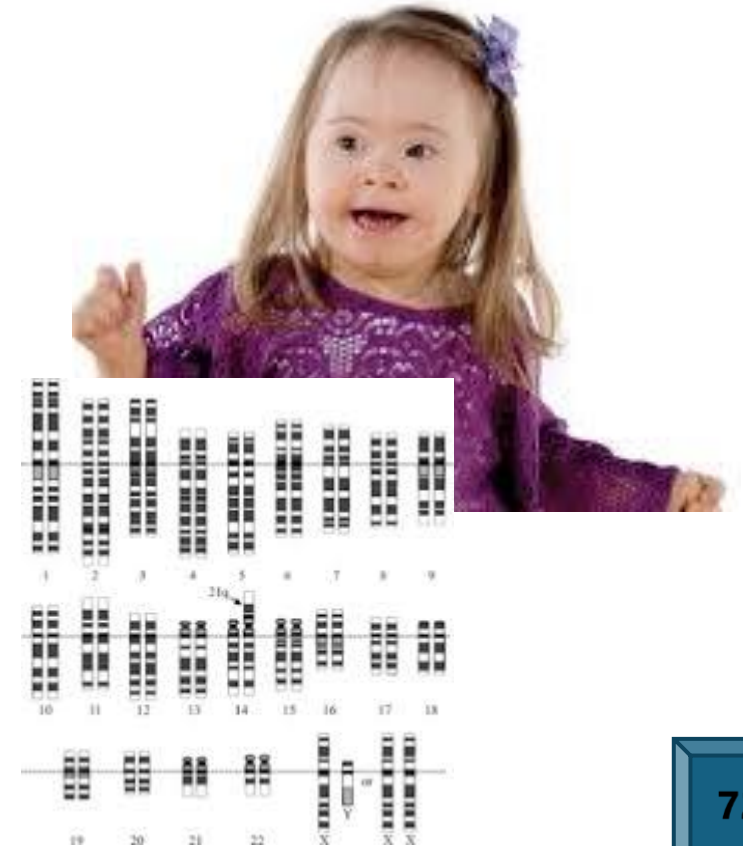
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Identify the different types of mutation that would result in the activation of a protooncogene to an oncogene such as point mutation, nonsense mutation, chromosomal translocation, and gene amplification; provide examples for each)

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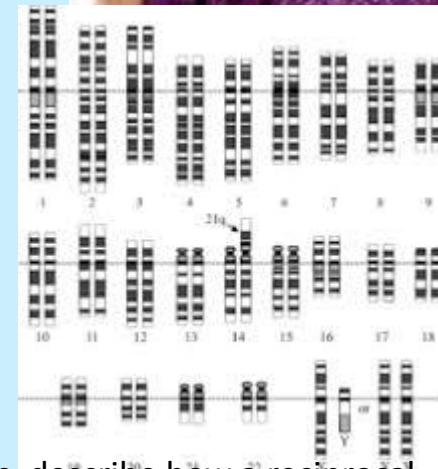
An 18-month-old girl is brought to the physician by her parents for a follow-up examination because she was born with a heart defect. Physical examination shows upslanted palpebral fissures, poor muscle tone, and a single palmar creases on both hands. Karyotype studies show 46 total chromosomes, which is interpreted as abnormal and causative of her condition. Which of the following best describes the aberration in this patient's karyotype?

- A. Maternal mitochondria 17%
- B. Robertsonian translocation 17%
- C. Unbalanced translocation 17%
- D. Pericentric inversion 17%
- E. Paracentric inversion 17%
- F. Ring chromosome 17%



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A 14-year-old boy is brought to the physician by his mother because of a history of progressive weight gain since childhood. At birth, he weighed 5 kg (11 lb), and at 6 weeks old his weight had increased to 8 kg (17.6 lb). After his 4th year of age, the weight gain progression became worse. He has had increased appetite to sugar-containing foods and has emotional outbursts when the meal size is reduced. He is 155 cm (5ft 1in) tall, weighs 160 kg (352 lb); BMI is 65 kg/m² (18.5-24.9 kg/m²). Physical examination shows a narrow nasal bridge, thin vermilion of the upper lip with downturned corners of the mouth. There are dry, dark patches on the skin of the neck and armpit, small testicles (1-2 ml) and no hair growth at the armpit or pubic area. Which of the following genetic aberrations is most likely in this patient?

- | | |
|---|-----|
| A. SNRPN gene deletion in paternal 15q11-13 | 20% |
| B. Reciprocal translocation between chr 4 and 8 | 20% |
| C. UBE3A gene deletion in maternal 15q11-13 | 20% |
| D. Robertsonian translocation between chr 14 and 21 | 20% |
| E. Microdeletion on the p arm of Chromosome 5 | 20% |

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A 65-year-old man comes to the physician because of a 3-week history of blood-stained sputum. Endoscopic studies of the bronchi show a tumor, and tissue samples are sent for genetic testing. Next-generation sequencing is performed on a panel of genes that include proto-oncogenes and tumor-suppressor genes. Results show a series of pathogenic mutations in a gene which functions as a 'guardian' of the genome. Which of the following genes most likely contains a pathogenic mutation?

- | | |
|----------------------|-----|
| A. Ras gene | 17% |
| B. Myc gene | 17% |
| C. TP53 gene | 17% |
| D. MLH1 gene | 17% |
| E. Gene encoding E2F | 17% |
| F. BRCA-1 gene | 17% |

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A 63-year-old man comes to the physician for his annual examination. A digital rectal examination shows a mass on the posterior aspect of the prostate gland. Several tissue biopsies are obtained. Some samples are immediately examined under the microscope and others are sent for Western blot analysis.

Microscopy shows findings consistent with prostate cancer, and Western blot analysis shows overexpression of the Bcl-2 protein in these cells . Which of the following best explains the mechanism by which this protein results in cancer progression in this patient?

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| A. Upregulates p53 | 20% |
| B. Prevents apoptosis | 20% |
| C. Prevents G1/S transition | 20% |
| D. Prevents caspase activation | 20% |
| E. Promotes ras activation | 20% |

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A 1-year-old boy is brought to the physician by his parents because of a 2-day history of developing freckling on his face and arms after a brief exposure to sunlight (image shown). Physical examination is consistent with the image. Targeted analysis of a panel of genes involved in DNA repair shows compound heterozygosity for pathogenic variants of the *XPC* gene. Which of the following processes is most likely to be defective in this patient?

- | | |
|--|-----|
| A. Activation of apoptosis | 17% |
| B. Nucleotide excision repair | 17% |
| C. Error in proof reading during replication | 17% |
| D. DNA mismatch repair | 17% |
| E. Repair of double stranded breaks in DNA | 17% |
| F. Base excision repair | 17% |



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A 4-year-old boy is brought to the physician by his mother because of a 1-year history of progressive ataxia, frequent respiratory infections and delayed motor milestones. His older sister has similar symptoms and was diagnosed with a genetic disorder characterized by defective DNA repair mechanisms. Physical examination shows visible telangiectasias on his face and conjunctiva. Genetic testing shows a mutation in the ATM gene. Which of the following best describes the normal function of the protein encoded by this gene?

- | | |
|--|-----|
| A. Detects double-strand breaks in DNA | 13% |
| B. Repairs double-strand breaks in DNA by non-homologous end joining | 13% |
| C. Repairs thymine dimers | 13% |
| D. Unwinds DNA during DNA replication | 13% |
| E. Nicks DNA by endonuclease activity | 13% |
| F. Mediates base excision repair | 13% |
| G. Mediates nucleotide excision repair | 13% |
| H. Proof-reads DNA during replication | 13% |

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- C. Repairs thymine dimers 13%
- D. Unwinds DNA during DNA replication 13%
- E. Nicks DNA by endonuclease activity 13%
- F. Mediates base excision repair 13%
- G. Mediates nucleotide excision repair 13%
- H. Proof-reads DNA during replication 13%

A 38-year-old man comes to the physician for evaluation of colon cancer because he has many relatives in his extended family with this condition. Colonoscopy shows a single large tumor in his colon. Genetic analysis shows that he, and the affected people in this family have hereditary nonpolyposis colon cancer. Which of the following best explains why this cancer is prevalent in many members of this family?

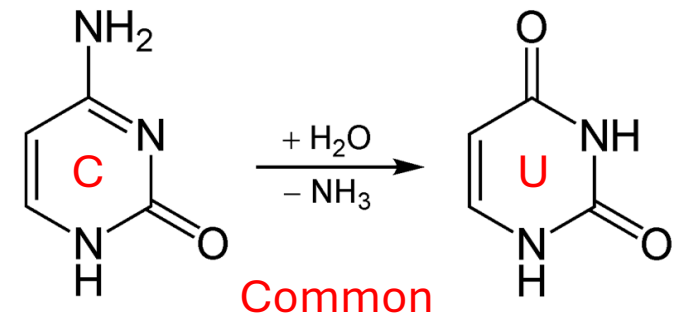
- | | |
|--|-----|
| A. Oncogene formation | 13% |
| B. Reduced DNA repair | 13% |
| C. Activation of apoptosis | 13% |
| D. Gene amplification | 13% |
| E. Activation of a cell cycle checkpoint | 13% |
| F. Formation of chromosome t(9;22) | 13% |
| G. Common environment | 13% |

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A 22-year-old woman comes to her physician for a follow-up appointment because of a 1-year history of recurrent infections. She has a history of multiple non-melanoma skin cancers and early-onset cataracts. Neurological examination shows balance difficulties and uncoordinated movements. Genetic testing shows a pathogenic variant in a gene encoding a protein required for a specific DNA repair mechanism. Laboratory studies shows an accumulation of deaminated cytosine residues in the DNA of her cells. Which of the following types of enzymes is most likely primarily responsible for repairing this type of DNA damage?

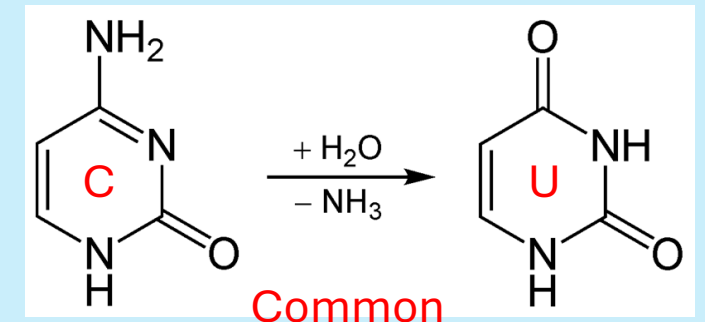
- | | |
|-------------------|----|
| A. Endonuclease | 0% |
| B. DNA polymerase | 0% |
| C. Glycosylase | 0% |
| D. Helicase | 0% |
| E. Ligase | 0% |



Uracil-DNA glycosylase will then perform base excision

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Uracil-DNA glycosylase will then perform base excision

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Analyze ways that DNA can be damaged, the cellular mechanisms that are employed in an attempt to fix the damage, and the long- and short-term consequences of mutation accumulation

An investigator is studying genome stability creates an assay to detect DNA repair by the proof-reading mechanism. The assay measures repair of mis-incorporated bases that were caused by tautomeric shifts of thymidine. In which of the following stages of the cell cycle did this repair most likely occur?

- A. G1 phase 0%
- B. S phase 0%
- C. G2 phase 0%
- D. M phase 0%
- E. G0 phase 0%

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A 35-year-old woman comes to the physician because of a 10-day history of a lump in her right breast. Physical examination shows a non-tender mass in the upper lateral side of the right breast that is 5 cm and is irregular. A biopsy was taken, and FISH analysis shows a breast carcinoma that is responsive to Herceptin treatment. Which of the following genetic mechanisms best explains her diagnosis?

- | | |
|----------------------------------|-----|
| A. BRCA-1 gene mutation | 17% |
| B. Chromosome 9:22 translocation | 17% |
| C. HER-2 overexpression | 17% |
| D. BRCA-2 gene mutation | 17% |
| E. Loss of function of P53 | 17% |
| F. Myc overexpression | 17% |

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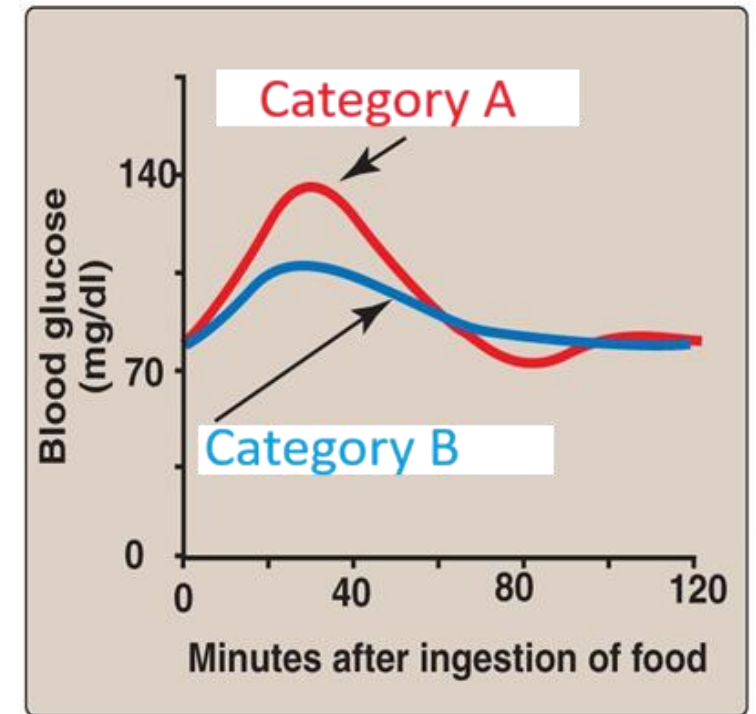
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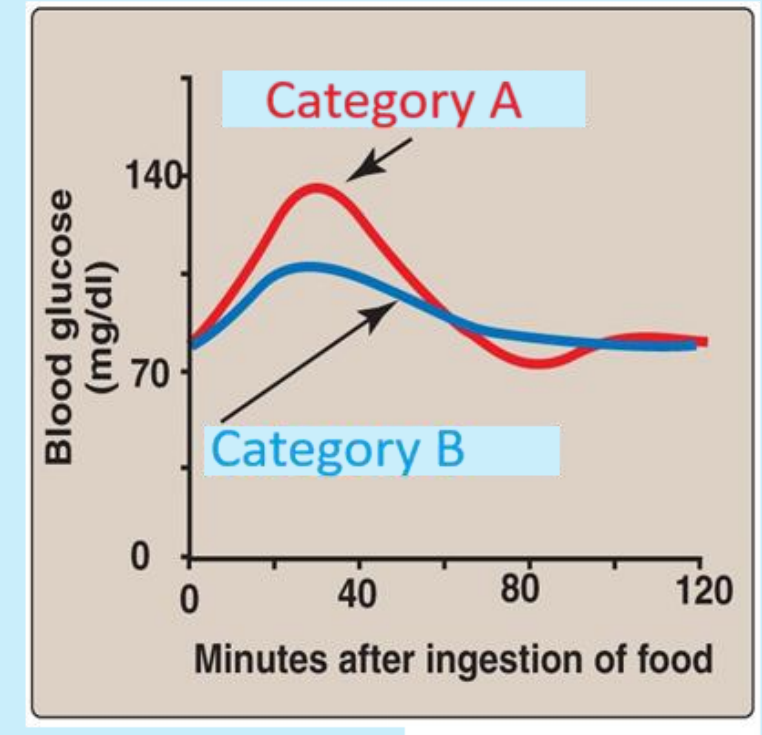
Biochemistry 4 Questions

A 52-year-old woman comes to the physician for follow-up after being diagnosed with type 2 diabetes mellitus one week ago. She was started on metformin and advised to make dietary changes. Her physician recommends that she adopts a diet which emphasizes food items belonging to category B on the postprandial blood glucose response curve as shown. Which of the following best explains the benefit of such a diet in this patient?



- | | |
|---|-----|
| A. Increased fluctuation of insulin release | 20% |
| B. Delayed glucose absorption | 20% |
| C. Decreased satiety | 20% |
| D. Increased overall caloric absorption | 20% |
| E. Increased frequency of hypoglycemic episodes | 20% |

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A 25-year-old man volunteers for a research study on appetite regulation. He is asked to fast overnight and then report to the research lab the next morning. Blood samples are taken at regular intervals. Blood tests in the morning show elevated levels of a peptide hormone. Which of the following best describes the source and function of this hormone?

- | | |
|---|-----|
| A. Adipose tissue; stimulates satiety | 20% |
| B. Duodenum; stimulates satiety | 20% |
| C. Stomach; stimulates hunger | 20% |
| D. Pancreas beta cells; stimulates hunger | 20% |
| E. Adrenal medulla; stimulates satiety | 20% |

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A 68-year-old woman comes to the clinic because of a 3-month history of progressive muscle weakness and unintended weight loss. She has a history of multiple hospitalizations for chronic illness and recent surgery. Physical examination shows muscle wasting and poor nutritional status. Which of the following physiologic findings best describes the nitrogen balance state in this patient?

- A. Nitrogen intake and excretion are at equilibrium 20%
- B. Increased urinary urea excretion 20%
- C. Increased protein synthesis 20%
- D. Decreased urea synthesis 20%
- E. Increased tissue growth 20%

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A 52-year-old man comes to the clinic to establish primary care. He has hypertension and type 2 diabetes mellitus, both well controlled on medications. His father had a myocardial infarction at age 55 years. He is 168 cm (5 ft 6 in) tall and weighs 82 kg (180 lb); BMI is 29 kg/m². Physical examination shows no abnormalities. Pulse is 88/min, and blood pressure is 138/94 mm Hg. Smoking cessation is recommended. Which of the following dietary changes is most likely to be recommended to this patient?

- A. Increased intake of saturated fatty acids 17%
- B. Increased intake of trans fatty acids 17%
- C. Increased intake of polyunsaturated fatty acids 17%
- D. Increased intake of dietary cholesterol 17%
- E. Decreased intake of fat-soluble vitamins 17%
- F. Increased intake of refined sugars 17%

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